

Class: Product						
Method	#	Test Description	Sample Input Data	Expected Output	Actual Output	P/F
Constructor	1	Create product with valid values	("Coke", 25.0, 10, Drinks, "Coca-Cola", "2025-12-01")	Object created correctly	Product added successfully	P
getName	2	Retrieve product name	Product("Coke")	"Coke"	Coke	P
setPrice	3	Update product price	setPrice(30)	Price = 30	Price = 30	P
reduceStock	4	Reduce stock by valid amount	qty=10, reduceStock(3)	qty = 7	Qty = 7	P
reduceStock	5	Reject invalid reduction	qty=10, reduceStock(20)	qty unchanged	Qty unchanged	P

Class: Inventory						
Method	#	Test Description	Sample Input	Expected Output	Actual Output	P/F
addProduct	1	Add new product	add(ProductX)	Product added	Product added successfully	P
findProductByName	2	Find existing product	"Coke"	Product returned	Product returned	P
restockProduct	3	Restock valid product	restock("Coke", 5)	qty increases	Coke has been restocked successfully	P
restockProduct	4	Restock nonexistent product	"Unknown"	"Not found"	Not found	P

Class: Customer						
Method	#	Description	Input	Expected	Actual	P/F
Constructor	1	Create customer	("Juan")	Customer created	Juan	P
isSenior	2	Check senior flag	senior=true	returns true	Returns true	P

Class: Cart						
Method	#	Test Description	Input	Expected	Actual	P/F
addItem	1	Add item to cart	add Coke x1	Added	As expected	P
updateQuantity	2	Update existing qty	set Coke=5	Updated	As expected	P
removeItem	3	Remove existing item	remove Coke	Removed	As expected	P
computeSubtotal	4	Compute correct subtotal	Coke=25 × 2	50.00	As expected	P

Class: Discount						
Method	#	Description	Input	Expected	Actual	P/F
computeSeniorDiscount	1	Compute 20%	100	20	20	P
computeVAT	2	Compute VAT 12%	100	12	12	P
computeTotal	3	Senior total	100, senior	80	80	P
computeTotal	4	Non-senior total	100	112	112	P

Class: Supplier						
Method	#	Description	Input	Expected	Actual	P/F
addProduct	1	Add valid product name	“Coke”	Added	Added	P
restockProduct	2	Restock valid product	Coke +5	qty increases	Qty increases	P
restockProduct	3	Restock product supplier doesn’t offer	“Unknown”	Fails	Does not restock	P

Class: Payment						
Method	#	Description	Input	Expected	Actual	P/F
Constructor	1	Create payment	100	Stored = 100	100	P
setAmount	2	Update amount	set 200	amount=200	Amount = 200	P

Class: Transaction						
Method	#	Description	Input	Expected	Actual	P/F
processTransaction	1	Non senior — VAT applied	subtotal=100	total=112	Total=112	P
processTransaction	2	Senior — no VAT	subtotal=100	total=80	Total=80	P
applyLoyaltyDiscount	3	Apply valid discount	discount=20	total=-20	Total=-20	P
finalizePayment	4	Exact payment	total=100 pay=100	change=0	Change=0	P
finalizePayment	5	Insufficient payment	total=100 pay=50	Insufficient payment. Amount Due: Php100	Insufficient payment. Amount Due: Php100	P

Class: Receipt						
Method	#	Description	Input	Expected	Actual	P/F
getFormattedReceipt	1	Generate receipt text	transaction	formatted text	Formatted text	P
saveToFile	2	Save receipt	fileName="receipt.txt"	File created	receipt.txt	P

Class: ProductCatalogPanel						
Method	#	Description	Input	Expected Output	Actual	P/F
refreshTable	1	Loads all products into table	inventory list	Table filled	Table w/ products	P
Add to Cart button	2	Add selected product	"Coke" qty=1	Cart updated	"Coke" Qty=1	P

Class: ShoppingCartPanel						
Method	#	Description	Input	Expected	Actual	P/F
refreshTable	1	Display cart items	cart with items	table updated	Updated table	P
Remove Item	2	Remove selected item	click remove	item removed	removes item	P
Update Qty	3	Update quantity	enter 5	qty=5	Qty = 5	P
Checkout	4	Go to checkout	click	moves panels	Moves to CheckoutPanel	P

Class: CheckoutPanel						
Method	#	Description	Input	Expected	Actual	P/F
refreshTotals	1	Compute non-senior totals	subtotal=100	total=112	Total = 112	P
refreshTotals	2	Compute senior totals	subtotal=100	total=80	Total = 80	P
processPurchase	3	Valid purchase	cash=200	payment success	Payment success	P
processPurchase	4	Not enough cash	cash < total	error message	Error message	P

Class: ReceiptPanel						
Method	#	Description	Input	Expected	Actual	P/F
Display receipt	1	Load receipt text		Show receipt	Shows receipt	P
Save Receipt	2	Click save button	click	File saved	File saved	P

Class: StoreGUI						
Method	#	Test Description	Sample Input Data	Expected Output	Actual Output	P/F
Constructor	1	Initialize GUI components	Run new StoreGUI()	Loads inventory, categories, panels, main menu	Loads inventory, categories, panels, main menu	P
createMainMenuPanel	2	Display main menu	Window opens	Buttons: Inventory, Transaction, Exit visible	Buttons: Inventory, Transaction, Exit visible	P
createInventoryPanel	3	Load inventory table	inventory contains products	Table displays product list	Table displays product list	P
showPage	4	Switch to Catalog page	showPage("Catalog")	Catalog panel becomes visible	Catalog panel becomes visible	P
showReceiptPanel	6	Display receipt after checkout	receipt + change	ReceiptPanel appears with text	ReceiptPanel appears with text	P
showReceiptPanel	7	Replaces old receipt panel	second receipt generated	Old panel removed, new one shown	Old panel removed, new one shown	P
main	8	Launch application	Run program	GUI window opens	GUI window opens	P